How PrusaSlicer and Bambu Studio turn a 3D model into G-code

A research report written before Caliper gets a slicer of its own. It explains, step by step, what the two slicers most people use actually do between "open a model" and "here is your G-code", using their real source code as the reference. Nothing in Caliper changes yet; this is the map to build from.

It is written for two readers at once:

- **A beginner engineer** who has never looked inside a slicer. Every term is explained the first time it appears, and each step comes with a picture in words, the real formula and a worked example with numbers.
- **A coding agent** that will later implement this. Every claim names the file and function it comes from, so it can be checked against the source instead of trusted.

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1. How to read this document

Versions. Everything here was read from these exact releases. Slicers move quickly, so if a later version disagrees, the later source wins.

Slicer	Release read	Where
PrusaSlicer	`version_2.9.6` (latest stable at the time of writing; `master` is a restructured 3.0 alpha)	`github.com/prusa3d/PrusaSlicer`, folder `src/libslic3r/`
Bambu Studio	`v02.08.04.61`	`github.com/bambulab/BambuStudio`, folder `src/libslic3r/`

References. A reference such as `PrintObject.cpp → PrintObject::prepare_infill` means: the file `src/libslic3r/PrintObject.cpp`, the function `PrintObject::prepare_infill`. Function names are given instead of line numbers because line numbers drift between releases.

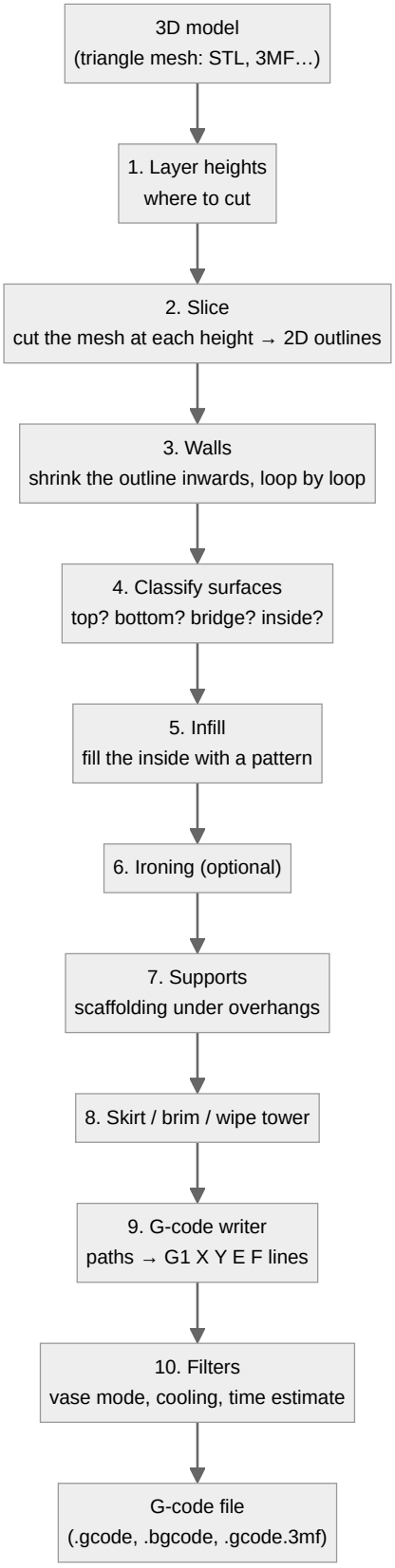
Two kinds of statements. When this report says what the code does, it was read from the source. When it explains *why*, that is the purpose stated in the source's comments or, where marked "in practice", general 3D-printing knowledge.

> **Warning**
>
> **Both slicers are licensed under the GNU AGPL v3. Caliper is MIT.** We may learn from their code and reimplement the *ideas and maths* in our own code, but we must not copy or translate their source into Caliper without changing Caliper's licence. Section 22.2 explains the options. This is the single most important constraint for the implementation work, so read it before writing any code.

2. The big picture in one page

A 3D printer cannot read a 3D model. It reads **G-code**: a text file of simple instructions such as "move the nozzle to X=10, Y=20 while pushing out 0.3 mm of filament". A **slicer** is the program that converts the model into those instructions.

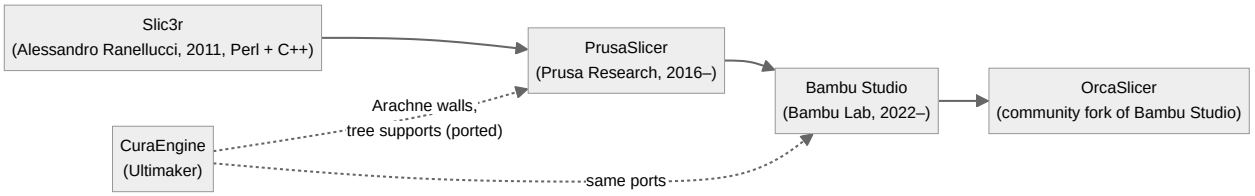
It does that by cutting the model into thin horizontal slices, like slicing a loaf of bread, and then deciding, for each slice, exactly where the nozzle should travel and how much plastic it should squeeze out along the way.



Steps 1–8 are **geometry**: they work on 2D polygons and produce lists of "extrusion paths" (lines with a width, a height and a role such as "outer wall"). Steps 9–10 are **G-code generation**: they turn those paths into text, choosing speeds, temperatures, fan, retractions and travel moves.

Both slicers run this same pipeline. Bambu Studio is a fork of PrusaSlicer, so the structure, and much of the code, is shared.

3. Where the two slicers come from



- **Slic3r** started the code base; it was partly written in Perl. Comments such as "ported from lib/Slic3r/GCode/CoolingBuffer.pm" still show this.
- **PrusaSlicer** rewrote most of it in C++ and is still where most of the core algorithms are developed. Its README says the geometry library is `libslic3r`.
- **Bambu Studio** forked PrusaSlicer. Its README says: "Bambu Studio is based on PrusaSlicer by Prusa Research, which is from Slic3r." It renamed many settings (`perimeters` became `wall_loops`), added features for Bambu printers (AMS filament changer, timelapse, printer-side checks) and packs its output into a `.gcode.3mf` file.
- Two important algorithms in both come from **CuraEngine** (Ultimaker's slicer): *Arachne* variable-width walls and *tree supports*. The source files keep Ultimaker's copyright lines.

Because of this history, when you understand one slicer you understand most of the other. This report describes the shared logic once and then lists the differences.

4. Words you need first

These come up on every page. A longer glossary is at the end.

Word	Meaning
Mesh	The 3D model as a list of triangles. An STL file is exactly this.
Layer	One horizontal slice of the print, e.g. 0.2 mm thick.
Polygon	A closed 2D shape made of straight edges, stored as a list of points.
ExPolygon	"Expanded polygon": one outer polygon (the *contour*) plus zero or more polygons cut out of it (the *holes*). A ring is one ExPolygon with one hole.
Offset	Moving every edge of a polygon outwards (positive) or inwards (negative) by a distance. Shrinking an outline by half a line width gives the centre line of the outer wall.
Boolean operations	Union (combine), difference (subtract), intersection (overlap) of polygons. The slicer uses them constantly. Both slicers use the **Clipper** library for this.
Perimeter / wall	A loop of plastic that follows the outline of the part. The outermost one is the *external perimeter*.
Infill	The pattern that fills the inside of the part. *Sparse* infill (e.g. 15%) is mostly air; *solid* infill (100%) is used for top and bottom skins.
Shell	The solid outside of the part: the walls plus the solid top and bottom layers.
Bridge	A line printed in mid-air between two supported points.
Overhang	Part of a layer that sticks out past the layer below it.
Extrusion	Pushing filament through the hot nozzle. In G-code it is the `E` value.
Travel	Moving the nozzle without extruding.
Retraction	Pulling the filament back a little before a travel so it does not ooze (drip) and leave strings.
Flow	How much plastic comes out per millimetre of movement, in mm^3/mm.
Feed rate	Movement speed. In G-code, `F` is in **mm per minute**, not per second.

5. The data model: what the slicer keeps in memory

Before the steps make sense you need to know what objects they read and write. The names below are the real C++ class names, identical in both slicers unless noted.

5.1 Input side: the model

```
Model                     everything on the plate
└── ModelObject           one object the user sees in the object list
    ├── ModelVolume       one mesh inside it (a part, a modifier, a negative volume…)
    │   └── TriangleMesh  the triangles
    └── ModelInstance     one copy of the object on the plate (position, rotation, scale)
```

An object can have several **volumes**: the main part, a *modifier* (a shape that changes settings inside it, e.g. more infill in one area), a *negative volume* (a shape subtracted from the part) or a *support blocker/enforcer*. An object can also have several **instances**, which are copies sliced once and printed several times.

5.2 Processing side: the print

```
Print                     the whole slicing job and its config
├── PrintObject           one ModelObject being sliced (shared by its instances)
│   ├── Layer[]           one per layer height
│   │   └── LayerRegion[] the part of this layer that uses one set of settings
│   │       ├── slices          ExPolygons: the cut outline, typed (top, bottom…)
│   │       ├── fill_surfaces   what is left for infill after the walls
│   │       ├── perimeters      the wall extrusion paths
│   │       └── fills           the infill extrusion paths
│   └── SupportLayer[]    support material, its own layers
└── PrintRegion[]         one per distinct set of region settings
```

A **region** is "the area of a layer that shares one set of settings". With no modifiers, every layer has one region. A modifier that sets 50% infill creates a second region for the area it covers.

5.3 Surface types

Every piece of a layer gets a **type**. This is how the slicer remembers "this part is a top surface" or "this part is mid-air". From `Surface.hpp → enum SurfaceType` (PrusaSlicer):

Type	Meaning
`stTop`	Seen from above. Printed solid and neat.
`stBottom`	Seen from below, resting on something. Printed solid.

Type	Meaning
`stBottomBridge`	Seen from below, resting on nothing: an overhang, printed with bridge flow.
`stInternal`	Inside the part. Gets sparse infill.
`stInternalSolid`	Inside, but must be solid (it supports a top surface or keeps a wall thick enough).
`stInternalBridge`	First solid layer printed over sparse infill: it bridges across the gaps in the infill.
`stInternalVoid`	Inside, deliberately left empty (e.g. combined infill layers).
`stPerimeter`	Taken by walls.
`stSolidOverBridge`	Internal solid directly above a bottom bridge (newer PrusaSlicer).

5.4 Extrusion paths and roles

The geometry steps end with **extrusion entities**: a polyline (list of points) plus `width`, `height`, `mm3_per_mm` (flow) and a **role**. The role tells the G-code writer which speed, acceleration and fan rules apply. Roles include external perimeter, perimeter, overhang perimeter, internal infill, solid infill, top solid infill, bridge infill, gap fill, ironing, skirt, support material, support interface and wipe tower.

5.5 Coordinates are integers

Floating-point numbers make polygon booleans unreliable, so both slicers store 2D coordinates as **64-bit integers** and scale millimetres up:

Slicer	`SCALING_FACTOR` (`libslic3r.h`)	1 integer unit =
PrusaSlicer	`0.000001`	1 nanometre (10^{-6} mm)
Bambu Studio	`0.00001`	10 nanometres (10^{-5} mm)

`scale_(mm) = mm / SCALING_FACTOR` and `unscale(units) = units * SCALING_FACTOR`. So in PrusaSlicer, 0.4 mm is stored as `400000`.

Both slicers are **Z-up**: X to the right, Y away from you, Z up, with the origin at the front-left corner of the bed in the final G-code.

6. The pipeline and its steps

6.1 The step lists

The pipeline is split into named steps. From `Print.hpp`:

Per object (`enum PrintObjectStep`):

PrusaSlicer 2.9.6	Bambu Studio 02.08.04.61
`posSlice`	`posSlice`
`posPerimeters`	`posPerimeters`
`posPrepareInfill`	`posPrepareInfill`
`posInfill`	`posInfill`
`posIroning`	`posIroning`
`posSupportSpotsSearch`	—
`posSupportMaterial`	`posSupportMaterial`
`posEstimateCurledExtrusions`	—
`posCalculateOverhangingPerimeters`	—
—	`posDetectOverhangsForLift`
—	`posSimplifyWall`, `posSimplifyInfill`, `posSimplifySupportPath`

Whole print (`enum PrintStep`): `psWipeTower` (also called `psToolOrdering`), `psSkirtBrim`, `psGCodeExport`. PrusaSlicer adds `psAlertWhenSupportsNeeded`; Bambu Studio adds `psConflictCheck`.

6.2 Why it is split into steps: invalidation

Each step records whether it has started and finished (`set_started` / `set_done`). When you change a setting, the slicer only throws away the steps that depend on it. Change the infill density and the walls are kept; change the layer height and everything from `posSlice` on is redone. Each step also **calls the step before it first**: `make_perimeters()` begins with `this->slice()`, `infill()` begins with `this->prepare_infill()`. That makes the steps safe to call in any order.

6.3 The order, as `Print::process` runs it

PrusaSlicer (`Print.cpp → Print::process`), simplified:

```
for each object, in parallel:
    make_perimeters()        # runs slice() first if needed
    infill()                 # runs prepare_infill() first if needed
    ironing()
for each object, one at a time:
    generate_support_spots() # finds spots that may need support, to warn the user
alert_when_supports_needed()
for each object, in parallel:
    generate_support_material()
    estimate_curled_extrusions()
    calculate_overhanging_perimeters()
make wipe tower, or just the tool ordering
make skirt and brim
check for collisions between objects' paths
```

The G-code export (`psGCodeExport`) runs afterwards, from `GCode.cpp → GCodeGenerator::do_export`.

Bambu Studio (`Print.cpp → Print::process`) has the same order with three additions:

1. **Shared objects.** Before slicing it compares objects (same mesh, same transform, same settings, same painting). Identical objects are sliced once and the result is copied (`set_shared_object`, `copy_layers_from_shared_object`). Ten identical parts slice as fast as one.
2. `detect_overhangs_for_lift()` runs after supports, so later travel moves can lift over overhangs.
3. Before the wipe tower it estimates print time per filament and checks which filaments a nozzle cannot physically reach (for multi-nozzle printers).

Parallelism. Both use Intel TBB (`tbb::parallel_for`) to process objects, and inside a step usually layers, on all CPU cores. A step that needs the layer below or above (for example classifying top surfaces) still runs in parallel, because it only *reads* neighbouring layers that the previous step already finished.

7. Step 1: choosing the layer heights

7.1 What it does

Before cutting, the slicer decides where every layer starts and ends. The result is a list of pairs `[bottom, top]`, one per layer.

7.2 The inputs

- `first_layer_height` (Bambu: `initial_layer_print_height`), e.g. 0.2 mm.
- `layer_height`, e.g. 0.2 mm.
- A **layer height profile**: a list of `(z, height)` pairs. With a fixed layer height it is just `[0, h, object_height, h]`. With *variable layer height* (the user painted it, or the adaptive algorithm in `SlicingAdaptive.cpp` chose it from the surface slope) it has many points.
- `min_layer_height` and `max_layer_height` from the printer. In PrusaSlicer, if the printer leaves them at 0, the minimum is `0.07` mm (`Slicing.cpp → MIN_LAYER_HEIGHT_DEFAULT`) and the maximum is `0.75 × nozzle diameter`.

7.3 The algorithm

From `Slicing.cpp → generate_object_layers` (the two slicers match):

```
out = []
print_z = 0
if the first layer has a fixed height:
    out += [0, first_layer_height]
    print_z = first_layer_height

loop:
    slice_z = print_z + min_layer_height / 2
    if slice_z >= object_height: stop
    height = look up the profile at slice_z,
             interpolating linearly between its two nearest points
    slice_z = print_z + height / 2
    if slice_z >= object_height: stop
    out += [print_z, print_z + height]
    print_z += height
```

Then `PrintObjectSlice.cpp → new_layers` creates one `Layer` per pair, with:

- `print_z` = the top of the layer (where the nozzle will be),
- `height` = top − bottom,
- `slice_z` **= the middle of the layer:** `(bottom + top) / 2`.

> **Why cut in the middle?** A layer of plastic stands for the whole slab between its bottom and its top. Cutting at the middle gives the average outline of that slab, so a sloped wall is wrong by at most half a layer instead of a whole one.

7.4 Worked example

A 1.0 mm tall part, first layer 0.2 mm, layer height 0.2 mm, minimum 0.07 mm:

Layer	Bottom	Top (`print_z`)	Cut at (`slice_z`)
0	0.0	0.2	0.1
1	0.2	0.4	0.3
2	0.4	0.6	0.5
3	0.6	0.8	0.7
4	0.8	1.0	0.9

After layer 4, `print_z` = 1.0, so the next `slice_z` would be 1.035, which is not below the object height of 1.0, so the loop stops. Five layers.

Note the source's own comment: `//FIXME Adjust the last layer to align with the top object layer exactly?`. If the height is not a multiple of the layer height, the top of the print can end slightly below or above the model's top.

8. Step 2: cutting the mesh into outlines

8.1 What it does

For every layer's `slice_z`, intersect the triangle mesh with a horizontal plane and get back the closed 2D outlines (ExPolygons) of the part at that height. This is `PrintObject::slice` → `PrintObject::slice_volumes`, which calls `TriangleMeshSlicer.cpp → slice_mesh_ex`.

8.2 Cutting one triangle

`TriangleMeshSlicer.cpp → slice_facet` handles one triangle and one plane:

- If all three corners are above or all below the plane, the triangle is not cut.
- Otherwise the plane crosses two of its edges. On each crossed edge between corners `a` and `b`, the crossing point is found by linear interpolation:

```
t = (slice_z - a.z) / (b.z - a.z)
point = a + t × (b - a)
```

- The two points form one **line segment** of the outline.
- The corners are reordered so the segment always has the same direction relative to the triangle's outward normal ("external on the right"). This keeps outer contours counter-clockwise and holes clockwise, which is how the slicer later tells a hole from a solid.
- Each segment remembers *which mesh edge or vertex* each end lies on. That is what makes the next step fast and exact.

Special cases decided in the source comments:

- A plane passing exactly through a corner is a "touch", not a cut.
- A horizontal triangle lying exactly in the plane: "an upward facing horizontal face is considered on slicing plane, while a downward facing horizontal face is considered not on slicing plane" (`TriangleMeshSlicer.hpp`).

8.3 Joining segments into loops

The segments of one layer are joined end to end into closed loops (`make_loops`):

1. **By connectivity** (`chain_lines_by_triangle_connectivity`). Two segments that share a mesh edge are neighbours, so they are chained without comparing coordinates at all. On a clean, closed ("manifold") mesh this closes every loop.
2. **Exact end points** (`chain_open_polylines_exact`). Leftover open chains are joined where their ends coincide, first in the same direction, then allowing reversed ones.

3. **Closing gaps** (`chain_open_polylines_close_gaps`). On a broken mesh, ends that are close but not equal are joined, longest chains first.

> **Why this matters for Caliper.** Caliper already reports a model's "open edges" (`src/viewer/printability.ts → openEdges`). Open edges are exactly what breaks step 1 and forces the slicer into the gap-closing step. A model that fails Caliper's closed-surface check may slice with missing or odd layers.

8.4 From loops to ExPolygons

`make_expolygons` turns the loops into ExPolygons using Clipper:

- **Fill rule.** The default mode `Regular` uses Clipper's *non-zero* rule: a point is inside if the loops around it wind around it a non-zero number of times. Overlapping shells in a sloppy model are merged instead of cancelling out. An `EvenOdd` mode exists for models that rely on it; `Positive` and `PositiveLargestContour` (used by vase mode) force all loops to count as solid (`MeshSlicingParams::SlicingMode`).
- **Closing radius** (`slice_closing_radius`). A tiny *morphological closing* (grow, then shrink by the same amount) fills hairline cracks.
- **Resolution** (`resolution`). Points closer together than this are merged, so a finely tessellated mesh does not produce millions of tiny segments.

8.5 Regions, painting and size compensation

After cutting, `slice_volumes` continues:

1. **Split into regions** (`slices_to_regions`). Modifier volumes and layer range settings decide which part of each layer belongs to which `PrintRegion`. Negative volumes are subtracted.
2. **Remove empty top layers.**
3. **Multi-material painting** (`apply_mm_segmentation`): faces the user painted with a different filament become their own regions. **Fuzzy skin painting** is handled the same way.
4. **XY size compensation** (`xy_size_compensation`): grow or shrink every outline by a fixed amount, to correct a printer that prints holes too small or parts too big. Positive values are applied while cutting, negative values here.
5. **Elephant foot compensation** (`elefant_foot_compensation`, spelled that way in the code): only on layer 0 and only when printing directly on the bed (no raft). The first layer is squashed and spreads outwards, so its outline is shrunk to compensate. `ElephantFootCompensation.cpp` shrinks it carefully, so narrow parts of the outline are not lost.
6. `Layer::make_slices` merges the regions into `lslices`, the islands of the whole layer, and orders them by a shortest path.

9. Step 3: walls (perimeters)

9.1 What it does

For each layer and region, turn the outline into nested loops of plastic, from the outside inwards, and work out what area is left for infill. This is `PrintObject::make_perimeters` → `Layer::make_perimeters` → `PerimeterGenerator.cpp`.

9.2 Extra perimeters on domed tops

Before generating walls, `PrintObject::make_perimeters` checks each layer against the one above. If the upper layer's outline falls into a narrow band just inside this layer's walls (the "critical area", 1.5 × perimeter spacing deep) along more than **30%** of its length, this layer gets one more wall, and the test repeats. This is for the top of a dome, where each layer is slightly smaller than the last and its wall would otherwise land on sparse infill. It is controlled by `extra_perimeters` and is skipped when there is no infill.

9.3 Classic walls

`PerimeterGenerator::process_classic` is the original Slic3r method:

```
last = this layer's outline
loop i = 0 .. wall_count-1:
    if i == 0:
        last = offset(last, −external_width / 2)        # centre line of the outer wall
    elif i == 1:
        last = offset(last, −(external_spacing + spacing) / 2)
    else:
        last = offset(last, −spacing)
    if nothing is left: stop
    store `last` as the centre line of wall i

inset   = (one wall) ? external_spacing / 2 : spacing / 2
overlap = infill_overlap (default 25%) of (inset + solid_infill_spacing / 2)
infill area = offset(last, −(inset − overlap)), with slivers narrower
              than one infill line removed

narrow leftovers between walls → "gap fill" (thin single lines)
parts too thin for even one wall → "thin walls" (if enabled)
```

Note that walls are stepped by **spacing**, not width (section 15.2): lines overlap a little so they fuse. The infill is pushed 25% *into* the last wall (`infill_overlap`) so the two bond instead of leaving a seam of air.

Every wall has the **same width**. If a part is 1.0 mm thick and each wall is 0.45 mm, two walls fit and a thin sliver is left over, which is either filled by a thin gap fill line or left empty.

9.4 Arachne walls (variable width)

`PerimeterGenerator::process_arachne` uses the `Arachne/` folder, ported from CuraEngine (`Arachne/WallToolPaths.hpp` keeps the Ultimaker copyright).

Instead of fixed-width offsets, Arachne computes the **medial axis** of the outline (the line through the middle of every thin feature, via a Voronoi diagram, a "skeletal trapezoidation") and then decides how many lines fit at each point and how wide each must be. Where a wall narrows, lines get thinner or merge; where it widens, a line is added. The result has fewer gaps and fewer tiny gap-fill lines, especially in text, thin ribs and sharp corners.

It is the C++ default in both (`perimeter_generator` / `wall_generator = Arachne`). The shipped profiles differ: PrusaSlicer's MK4S profiles use `arachne`, while Bambu Studio's base profile (`fdm_process_common.json`) sets `wall_generator = classic`.

9.5 Wall order

Both can print outer-then-inner or inner-then-outer.

- Inner first (the default in both) gives the outer wall something to press against, which looks better and handles overhangs better.
- Outer first gives more accurate outer dimensions.

PrusaSlicer: `external_perimeters_first` (default off). Bambu Studio: `wall_sequence` with `InnerOuter` (default), `OuterInner` and `InnerOuterInner` (outer wall printed between two inner ones).

9.6 The seam

A wall is a closed loop, so it must start and end somewhere. That point leaves a small mark: the **seam**. Both have `seam_position`:

Option	Behaviour
Aligned (default in both)	Seams line up vertically, preferring hidden concave corners.
Nearest	Start wherever is closest to the nozzle. Fast, scattered seams.
Rear	Put the seam at the back of the part.
Random	Scatter seams to avoid a visible line.

PrusaSlicer 2.9 rewrote the seam placer into several files (`GCode/SeamPlacer.cpp`, `SeamAligned.cpp`, `SeamRear.cpp`, `SeamRandom.cpp`, `SeamScarf.cpp`, …). Bambu Studio keeps the earlier single `SeamPlacer.cpp`. Both support **scarf seams**: the loop's start and end ramp up and down in Z and flow so they overlap smoothly instead of butting together.

10. Step 4: preparing infill: classifying every surface

This is the step that decides "solid here, sparse there, bridge there". It is `PrintObject::prepare_infill`, and it runs these sub-steps in this order.

10.1 `detect_surfaces_type`: top, bottom or inside?

For every layer, compare its outline with the layers directly above and below (PrusaSlicer, `PrintObject::detect_surfaces_type`):

```
top    = this layer − layer above      (what nothing covers)
bottom = this layer − layer below      (what nothing holds up)
inside = everything else
```

`−` is a polygon difference. Each result is cleaned with a small *opening* (shrink then grow by 1/10 of the external perimeter width) to delete slivers too thin to matter.

The top layer of the object is all `stTop`. Bottom areas become `stBottomBridge` (an overhang, bridged in mid-air) unless the layer is printed on soluble support touching the part (`support_material_contact_distance = 0`), in which case it is `stBottom`.

Picture a mug: the bottom disc is `stBottom` on layer 0. The inside of the mug's floor is `stTop` on the last floor layer, because nothing is printed above it. The handle's underside where it leaves the body is `stBottomBridge`.

10.2 `prepare_fill_surfaces`

Turns top/bottom into plain internal if the user set zero top or bottom layers, and turns tiny internal areas into solid ones, where sparse infill would be a few pointless strokes.

10.3 `discover_vertical_shells`: keep the walls thick on slopes

On a gentle slope, each layer's walls sit mostly on the infill below, so the shell would be thin. This step adds solid infill next to sloped walls so the shell keeps the thickness the top/bottom layer counts promise. In PrusaSlicer it follows `ensure_vertical_shell_thickness`, which in 2.9 is a three-way choice (disabled, partial, enabled; enabled by default) rather than a switch.

10.4 `process_external_surfaces`: bridges

From the source comments:

1. Bottom and bottom-bridge areas are grown by 3 mm into the infill area, so a bridge is anchored into solid material on both sides. **Bridge direction** is chosen here (`BridgeDetector.cpp`): the direction in which the lines are best supported at both ends.
2. Top areas are grown by 3 mm the same way.
3. Internal areas are clipped by the grown areas.

4. Surfaces of the same kind are merged.

10.5 `discover_horizontal_shells`: top and bottom skins

A single layer marked `stTop` is not enough; the user asked for, say, 5 top layers. This step walks *down* from every top surface for `top_solid_layers − 1` layers (Bambu: `top_shell_layers`) and *up* from every bottom surface for `bottom_solid_layers − 1` layers, marking those areas `stInternalSolid`. Both slicers also have a minimum shell *thickness* option (`top_solid_min_thickness` / `top_shell_thickness`), which adds layers when the layers are very thin.

10.6 `bridge_over_infill`

The first solid layer above sparse infill would sag into the infill's gaps. This step finds those areas and makes them `stInternalBridge`: printed with bridge flow and speed, in a direction chosen to cross the infill lines.

10.7 `combine_infill`

If `infill_every_layers` is N > 1, N layers of sparse infill are printed as one thicker layer, every N-th layer, and the others become `stInternalVoid`. This saves time when the nozzle can lay down thicker lines.

10.8 Infill above bridges (PrusaSlicer)

Areas of solid infill directly above bridges are split off (`InfillAboveBridges.cpp → separate_infill_above_bridges`) so they can use `over_bridge_speed`.

11. Step 5: infill patterns

`PrintObject::infill` → `Layer::make_fills` (`Fill/Fill.cpp`).

11.1 Grouping

`group_fills` collects all fill surfaces of a layer that share the same pattern, density, angle, flow and bridge state, so each group is filled in one go. Top surfaces get the top pattern, bottoms the bottom pattern, internal solid the solid pattern and internal sparse the sparse pattern.

11.2 Line spacing from density

For line-based patterns (`Fill/FillRectilinear.cpp`):

```
line_spacing = spacing / density
```

`spacing` is the flow's centre-to-centre distance (section 15). At 100% the lines touch; at 20%, `0.45 / 0.20 ≈ 2.25 mm` apart, roughly.

11.3 Angle

`fill_angle` (Bambu: `infill_direction`) defaults to **45°**. The base class rotates every other layer by 90° (`FillBase.hpp → _layer_angle` returns 0 or π/2 depending on the layer index), so consecutive layers cross. Bridges ignore this and use the direction found by the bridge detector.

11.4 The patterns

PrusaSlicer 2.9.6 (`enum InfillPattern`): rectilinear, monotonic, monotonic lines, aligned rectilinear, grid, triangles, stars, cubic, line, concentric, honeycomb, 3D honeycomb, gyroid, Hilbert curve, Archimedean chords, octagram spiral, adaptive cubic, support cubic, support base, lightning, ensuring, zig-zag.

Bambu Studio adds: cross hatch, cross zag, locked zag, floating concentric, 2D lattice, an internal concentric, an Archimedean-spiral ironing pattern and a global monotonic line.

What the important ones are:

Pattern	Idea	Typical use
Rectilinear	Parallel lines, alternating 90° per layer.	Solid layers.
Monotonic	Rectilinear, but every line is printed in the same direction, side to side. No visible "stitch" lines on top.	Top surfaces.
Grid	Lines in two directions on every layer.	Fast sparse infill (Bambu and Prusa profiles both ship it).

Pattern	Idea	Typical use
Cubic / adaptive cubic	Lines that form 3D cubes standing on a corner; adaptive cubic makes the cubes bigger deep inside the part, where strength matters less.	Strong sparse infill / saving material.
Gyroid	A smooth wavy 3D surface; equally strong in all directions.	Strong sparse infill.
Lightning	Tree-like infill that only supports the top surfaces.	Fast, weak parts.
Concentric	Rings following the outline.	Flexible parts, round tops.

Patterns such as gyroid and cubic are **3D**: the 2D lines on a layer depend on the layer's Z, so the layers stack into a 3D structure.

11.5 Anchors

Infill lines end at the inner wall. "Anchoring" (`infill_anchor` / `sparse_infill_anchor`) connects line ends along the wall so lines are fastened instead of left dangling. Bambu's default is `400%` of the line width.

12. Step 6: ironing

Optional (`ironing` / `ironing_type`). After a top surface is printed, the nozzle passes over it again with a very small flow, melting and smoothing it. Choices: top surfaces, only the topmost surface, or all solid surfaces. Implemented in `PrintObject::ironing` / `Layer::make_ironing`.

13. Step 7: supports

13.1 Finding what needs support

An overhang needs support if it sticks out too far past the layer below.

PrusaSlicer (`Support/SupportMaterial.cpp`, function computing the overhangs):

```
if support_material_threshold == 0 (automatic):
    allowed overhang = 0.5 × external perimeter width
else:
    allowed overhang = layer height / tan(threshold angle + 1°)
needs support = this layer − (layer below grown by allowed overhang)
```

The threshold angle is measured **from the horizontal** (PrusaSlicer's tooltip: "90° = vertical"). A 45° threshold allows each layer to step out by one layer height.

Bambu Studio uses `support_threshold_angle`, default **30°**, also from the horizontal: "Support will be generated for overhangs whose slope angle is below the threshold."

> **Caliper note.** Caliper's overhang highlight (`src/viewer/printability.ts`) measures the limit **from vertical**, like Bambu Studio's "overhang threshold" in its *Preview*. Converting between the two: `from_vertical = 90° − from_horizontal`.

13.2 Styles

PrusaSlicer `support_material_style`	**Bambu Studio** `support_type` + `support_style`
`grid` (default), `snug`, `organic` (and `tree`, internally)	type: `normal(auto)` (default), `tree(auto)`, `normal(manual)`, `tree(manual)`; style: default, grid, snug, tree slim, tree strong, tree hybrid, organic

- **Normal / grid supports** project the overhang straight down to the bed or the part and fill the column with a sparse pattern.
- **Snug** follows the part's outline more closely.
- **Tree / organic supports** grow branches that start at a few points on the bed and split as they rise, only touching the part where needed. Both slicers' organic supports are Thomas Rahm's tree supports, "loosely based on Tree Supports by CuraEngine" (`TreeSupport.cpp`, Bambu's `TreeSupport3D.cpp`). Bambu also has its own older tree support (`TreeSupport.cpp`) for the slim, strong and hybrid styles.

13.3 Interfaces and gaps

- **Contact (Z) distance** (`support_material_contact_distance`, default 0.2 mm; Bambu `top_z_distance`): the air gap between the support's top and the part, so the support breaks away.
- **Interface layers**: denser layers at the top of the support, giving the part a flatter surface to rest on.
- **Raft**: a few layers printed under the whole part.
- Support **enforcers** and **blockers** (painted, or modifier volumes) force or forbid support in an area.

14. Step 8: skirt, brim, wipe tower and tool order

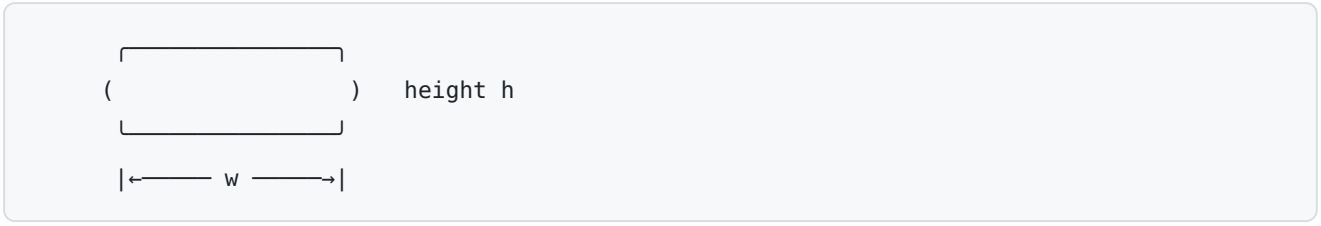
- **Skirt**: one or more loops around the parts, not touching them, to prime the nozzle. With a *draft shield* the skirt is printed all the way up as a wall.
- **Brim**: loops attached to the first layer of the part, to stop corners lifting. Bambu Studio also has *brim ears*: small brim patches only at sharp corners.
- **Tool ordering** (`GCode/ToolOrdering.cpp`): for multi-material prints, decides which filaments are used on each layer and in which order, to minimise filament changes.
- **Wipe tower / prime tower** (`GCode/WipeTower.cpp`): a tower beside the parts where the old filament is purged after each change. How much is purged depends on the pair of colours: PrusaSlicer's `wiping_volumes_matrix`, Bambu Studio's `flush_volumes_matrix` (dark-to-light needs more purging than light-to-dark).

15. The flow maths: how much plastic each line needs

This is the heart of "G-code conversion". If you remember one section, make it this one.

15.1 The shape of an extruded line

A line of molten plastic squashed onto the layer below is not a rectangle. Both slicers model it as a **rectangle with semicircles on both sides** ("rounded rectangle"), `width` wide and `height` tall:



15.2 The formulas

From `Flow.cpp` (identical in both slicers):

```
cross_section_area (mm²)  = h × (w − h × (1 − π/4))        # mm3_per_mm()
spacing (mm)              = w − h × (1 − π/4)              # centre-to-centre distance
bridge area (mm²)         = π × w² / 4                     # a round thread in mid-air
bridge spacing (mm)       = w + 0.05                       # BRIDGE_EXTRA_SPACING
```

`1 − π/4 ≈ 0.2146`. The rounded sides of neighbouring lines overlap a little, which is why lines are placed `spacing` apart rather than `width` apart.

Default widths if the user leaves a width at 0 (`Flow.cpp → auto_extrusion_width`):

Role	Width
Support, support interface, top solid infill	1.0 × nozzle diameter
Everything else (walls, infill, solid infill)	1.125 × nozzle diameter

For a 0.4 mm nozzle that is 0.4 mm and 0.45 mm. In practice the shipped profiles set explicit widths: PrusaSlicer's MK4S profile uses 0.45 mm (0.42 mm for top infill); Bambu's base profile uses 0.42 mm for the outer wall and top surface and 0.45 mm for inner walls and sparse infill.

15.3 From volume to filament length: the E value

The printer does not know about volume; it moves the filament by a length `E`. From `Extruder.cpp` and `GCode.cpp`:

```
filament_area   = π × (filament_diameter / 2)²
e_per_mm3       = flow_multiplier / filament_area
e_per_mm        = e_per_mm3 × cross_section_area
E for a segment = e_per_mm × segment_length
```

- `flow_multiplier` is `extrusion_multiplier` in PrusaSlicer and `filament_flow_ratio` in Bambu Studio. It corrects for a filament that comes out slightly over or under.
- PrusaSlicer has `use_volumetric_e`: if on, it does **not** divide by the filament area and writes E in mm³ (some firmware wants that). Bambu Studio always divides.

15.4 Worked example

A 0.4 mm nozzle, 1.75 mm filament, flow multiplier 1.0, a 0.2 mm layer, a 0.45 mm wide wall line, 10 mm long:

```
cross_section = 0.2 × (0.45 − 0.2 × 0.2146)
              = 0.2 × (0.45 − 0.04292)
              = 0.2 × 0.40708
              = 0.081416 mm²          (= mm³ of plastic per mm of line)

filament_area = π × 0.875² = 2.40528 mm²

e_per_mm      = 0.081416 / 2.40528 = 0.033849

E for 10 mm   = 0.33849 mm of filament
```

So the G-code line is roughly `G1 X… Y… E0.33849` (relative E) and the line is spaced `0.45 − 0.04292 = 0.40708 mm` from its neighbour.

> The number `2.40528` appears in Bambu's own printer G-code templates as `2.4053`: they divide a volumetric speed (mm³/s) by it to get a filament feed rate (mm/s). Now you know where it comes from.

15.5 The volumetric speed cap

A hot end can only melt so much plastic per second. The slicer therefore caps speed:

```
max_speed = max_volumetric_speed / cross_section_area
speed     = min(requested_speed, max_speed)
```

With a 15 mm³/s limit and the 0.081416 mm² line above, the cap is `15 / 0.081416 ≈ 184 mm/s`. Ask for 250 mm/s and you get 184. In PrusaSlicer this happens in `cap_speed` inside `GCodeGenerator::_extrude`, "even if user is not using autospeed". If a role's speed is set to 0, PrusaSlicer uses the volumetric limit itself as the speed (`m_volumetric_speed / mm3_per_mm`).

16. Step 9: writing the G-code

16.1 The file, top to bottom

PrusaSlicer, `GCode.cpp → GCodeGenerator::_do_export`:

1. Header comment (`; generated by PrusaSlicer …`).
2. Thumbnails (PNG/QOI/JPG, base64 in comments), if the printer wants them.
3. The print's notes and a summary of extrusion widths, as comments.
4. A placeholder for the first progress line (`M73`), filled in later.
5. Machine limits (`M201`, `M203`, `M204`, `M205`) if configured.
6. Object labels, so the printer can cancel one object mid-print.
7. First-layer bed, chamber and nozzle temperatures, **unless the custom start G-code already sets them**.
8. **The custom start G-code**, with placeholders filled in (section 16.6).
9. Preamble: `G21` (mm), `G90` (absolute XY), `M82`/`M83` (absolute or relative E).
10. The first tool selection.
11. **Every layer** (16.2), in a parallel pipeline (17.1).
12. The custom end G-code.
13. Statistics (filament used, time estimates) and the full config as comments.

16.2 Inside one layer

`GCodeGenerator::process_layer`, for all objects at one `print_z`:

1. Custom `before_layer_gcode`.
2. **Layer change** (`change_layer`): move Z, retract first if `retract_layer_change` is on.
3. Custom `layer_gcode`.
4. On the second layer: switch from first-layer to normal temperatures, and set the normal bed temperature.
5. A colour change or pause the user placed at this height (`M600` and similar).
6. **For each extruder used on this layer**, in tool order:
    1. Tool change (through the wipe tower if there is one).
    2. Skirt, then brim (first layer).
    3. Support.
    4. For each object instance, for each island: **walls, then infill**, or infill first if `infill_first` is set.
    5. Ironing.

16.3 One extrusion path → G-code lines

`GCodeGenerator::_extrude`, for each path:

1. **Travel** to the path's start (16.4), then **unretract** (push the filament back).

2. **Acceleration** by role, first match wins: first layer, first layer over raft, bridge, top solid infill, solid infill, infill, external perimeter, perimeter, default.
3. **e_per_mm** from the flow maths (15.3).
4. **Speed** by role: `perimeter_speed`, `external_perimeter_speed`, `bridge_speed`, `infill_speed`, `solid_infill_speed`, `over_bridge_speed`, `top_solid_infill_speed`, `ironing_speed`, `gap_fill_speed`.
5. **First layer** overrides: `first_layer_speed` (and `first_layer_infill_speed` for solid infill).
6. **Overhang slow-down** (16.5).
7. **Volumetric cap** (15.5).
8. `F = speed × 60` (mm/min).
9. Comment tags for the later filters: `;TYPE:…` (the role), `;WIDTH:…`, `;HEIGHT:…`, and cooling markers such as `;_EXTRUDE_SET_SPEED`.
10. For each segment: `G1 X Y E` (or `G2`/`G3` arcs if arc fitting is on; or `G1 X Y Z E` where Z changes mid-path, for scarf seams and vase mode). `E = e_per_mm × segment_length × e_fraction` (`e_fraction` lets a path ramp its flow, for scarf seams).

Coordinates are rounded ("quantized") to the output precision before writing, and a segment that rounds to zero length is skipped.

16.4 Travel, retraction, wipe and Z-hop

When moving between two paths:

- **Retract or not?** (`GCodeGenerator::needs_retraction`) Do not retract if the travel is shorter than `retract_before_travel` (Bambu: `retraction_minimum_travel`, both 2 mm by default), if it stays inside one support island, or (with `only_retract_when_crossing_perimeters`) if it stays inside the part over infill, where a string would be hidden. Otherwise retract.
- **Retraction**: `G1 E-<length>` at retraction speed. Defaults: PrusaSlicer 2 mm, Bambu 0.8 mm (direct-drive extruders need less).
- **Wipe**: optionally move along the just-printed path while retracting, to wipe ooze off the nozzle.
- **Z-hop / lift**: lift the nozzle during travel so it does not drag across the part (`retract_lift`; Bambu `z_hop`, default 0.4 mm, with types normal, slope and spiral; spiral is Bambu's default).
- **Avoid crossing perimeters** (`avoid_crossing_perimeters`; Bambu `reduce_crossing_wall`): route travels around the inside of walls instead of across them, so ooze stays inside.

16.5 Slowing down on overhangs

Both slow down on walls that overhang, but compute it differently.

PrusaSlicer (`GCode/ExtrusionProcessor.cpp → calculate_overhang_speed`) is continuous. For each point of an overhanging wall it measures how far the line sticks out past the layer below (0 = fully supported, one line width = fully in the air). The user sets speeds `overhang_speed_0..3` for 0%,

25%, 50% and 75% overhang, and the speed is interpolated linearly between them. The same mechanism raises the fan (`overhang_fan_speed_0..3`). It is controlled by `enable_dynamic_overhang_speeds`.

Bambu Studio (`GCode.cpp → get_overhang_degree_corr_speed`) uses buckets. Each overhanging path gets an `overhang_degree` from 0 to 5:

Degree	How much of the line overhangs	Setting
1	10–25%	`overhang_1_4_speed`
2	25–50%	`overhang_2_4_speed`
3	50–75%	`overhang_3_4_speed`
4	75–100%	`overhang_4_4_speed`
5	100% (no contact)	`overhang_totally_speed`

For a fractional degree below 4 it interpolates between the two neighbouring bucket speeds:

```
speed = speed[floor(d)] + (speed[floor(d)+1] − speed[floor(d)]) × (d − floor(d))
```

where a 0 setting means "use the normal speed". From degree 4 on it uses the bucket speed directly ("use lower speed of 75%-100% for better cooling"). Bambu also smooths sudden speed jumps along a wall (`smooth_speed_discontinuity_area`) in steps of 10 mm/s.

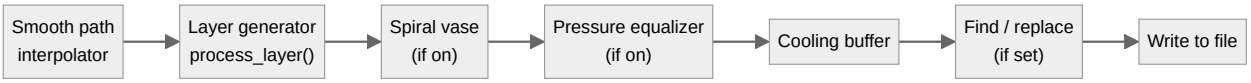
16.6 Custom G-code and placeholders

Start, end, layer-change and tool-change G-code are templates written by the printer profile. The **PlaceholderParser** (`PlaceholderParser.cpp`) fills them in: `[first_layer_temperature]`, `{layer_num}`, and a small language with `{if …}{elsif …}{else}{endif}` and arithmetic. Bambu's printer templates use it heavily (section 19.3).

17. Step 10: the post-processing filters

17.1 The per-layer pipeline

The layers are not written one by one. PrusaSlicer runs a **pipeline** of filters (`GCodeGenerator::process_layers`, a `tbb::parallel_pipeline` with up to 12 layers in flight). Each layer's G-code passes through, in order:



`serial_in_order` means each filter sees the layers in order, while different layers are at different stages at the same time.

17.2 Spiral vase

`GCode/SpiralVase.cpp`. In vase mode the part is a single wall printed as one continuous spiral: instead of stepping Z at each layer, Z rises smoothly along each loop, so there is no seam. Only possible for a single object with a single wall and no top layers.

17.3 Pressure equalizer

`GCode/PressureEqualizer.cpp`. "Finds changes in the volumetric extrusion speed and adjusts the transitions." A sudden jump from slow to fast extrusion under-extrudes for a moment because pressure in the nozzle lags behind. This filter ramps the speed so the volumetric rate changes no faster than `max_volumetric_extrusion_rate_slope_positive/negative` (0 = off by default).

17.4 Cooling buffer: slow down small layers, set the fan

`GCode/CoolingBuffer.cpp`. A small layer prints so quickly that it is still soft when the next layer goes on top. This filter collects a whole layer, measures how long it takes, and fixes that.

Slow-down. If the layer would take less than `slowdown_below_layer_time` (Bambu: `slow_down_layer_time`, default 5 s), the adjustable moves are stretched so the layer takes that long, but no move goes below `min_print_speed` (Bambu: `slow_down_min_speed`, default 10 mm/s). External walls and similar are slowed less, or last, so the visible surface keeps its quality.

Fan. From `CoolingBuffer.cpp → apply_layer_cooldown`:

```
if layer_time < slowdown_below_layer_time:
    fan = max_fan_speed
elif layer_time < fan_below_layer_time:
    t   = (layer_time − slowdown_below_layer_time)
          / (fan_below_layer_time − slowdown_below_layer_time)
    fan = t × min_fan_speed + (1 − t) × max_fan_speed
else:
    fan = min_fan_speed if fan_always_on else 0
```

The fan is off for the first `disable_fan_first_layers` layers and ramps up to full by `full_fan_speed_layer`. Bridges use `bridge_fan_speed`.

Example (PrusaSlicer's C++ defaults: slow down below 5 s, fan below 60 s, min 35%, max 100%), a 30 s layer:

```
t   = (30 − 5) / (60 − 5) = 0.4545
fan = 0.4545 × 35 + 0.5455 × 100 = 15.9 + 54.5 = 70.45  →  70%
```

Bambu Studio uses the same formula, moved to `GCode/GCodeEditor.cpp`, with its names: `slow_down_layer_time`, `fan_cooling_layer_time`, `fan_min_speed` (default 20%), `fan_max_speed`.

17.5 GCodeProcessor: time and filament estimates

`GCode/GCodeProcessor.cpp` reads the finished G-code back like a printer would, simulating acceleration (trapezoid speed profiles) and the machine limits, to estimate the print time. It then fills in placeholders it left earlier, e.g. `M73 P<percent> R<minutes left>` progress lines, and computes filament length, volume, weight and cost per extruder. PrusaSlicer computes two estimates: normal mode (`M73 P… R…`) and Prusa's quieter "stealth" mode (`M73 Q… S…`). This is also what powers the G-code preview in both apps.

18. G-code commands you will see

The common subset (Marlin-style, which both use by default for their own printers):

Command	Meaning	Example
`G0` / `G1`	Move in a straight line; with `E`, extrude while moving. `F` is speed in mm/min.	`G1 X50 Y40 E0.338 F9000`
`G2` / `G3`	Arc, clockwise / counter-clockwise (arc fitting). `I` / `J` = centre offset.	`G2 X10 Y10 I5 J0 E0.5`
`G28`	Home the axes.	`G28`
`G90` / `G91`	Absolute / relative XYZ coordinates.	`G90`
`G92`	Set the current position without moving (often `G92 E0` to reset E).	`G92 E0`
`M82` / `M83`	Absolute / relative E. Bambu's default is relative (`use_relative_e_distances` = true); PrusaSlicer's C++ default is absolute, though its Prusa printer profiles use relative.	`M83`
`M104` / `M109`	Set nozzle temperature / set and **wait**.	`M109 S220`
`M140` / `M190`	Set bed temperature / set and wait.	`M140 S60`
`M106` / `M107`	Fan on (S0–255) / off. Bambu uses `P1`, `P2`, `P3` for the part, auxiliary and chamber fans.	`M106 S178`
`M204`	Acceleration.	`M204 S5000`
`M73`	Progress: `P` percent, `R` minutes left. Bambu also uses `L` for the layer.	`M73 P42 R17`
`T<n>`	Select tool / extruder.	`T1`
`M600`	Filament change (colour change).	`M600`
`M486` / `EXCLUDE_OBJECT_*`	Label objects so one can be cancelled (Marlin/Prusa vs Klipper syntax).	`M486 S0`

Comment tags that both slicers' previews rely on: `;LAYER_CHANGE`, `;Z:`, `;HEIGHT:`, `;TYPE:External perimeter`, `;WIDTH:`.

A small annotated example

This is **illustrative and simplified**, not copied from a real file, to show how the pieces fit:

```
;LAYER_CHANGE
;Z:0.4
;HEIGHT:0.2
G1 E-0.8 F2100           ; retract before moving up
G1 Z0.4 F720             ; layer change
G1 X95.2 Y88.1 F12000    ; travel to the start of the outer wall
G1 E0.8 F2100            ; unretract
;TYPE:External perimeter
;WIDTH:0.42
G1 F7200                 ; 120 mm/s × 60
G1 X104.8 Y88.1 E0.30100  ; 9.6 mm × 0.031354 E/mm (0.42 wide, 0.2 high, 1.75 mm filament)
G1 X104.8 Y97.7 E0.30100
...
```

19. What Bambu Studio does differently

19.1 Renamed settings

Bambu Studio renamed most settings. When reading its code next to PrusaSlicer's, this table saves a lot of searching:

Concept	PrusaSlicer	Bambu Studio
Number of walls	`perimeters`	`wall_loops`
Wall generator	`perimeter_generator`	`wall_generator`
Wall order	`external_perimeters_first`	`wall_sequence`
Top / bottom solid layers	`top_solid_layers` / `bottom_solid_layers`	`top_shell_layers` / `bottom_shell_layers`
Sparse infill pattern	`fill_pattern`	`sparse_infill_pattern`
Sparse infill density	`fill_density`	`sparse_infill_density`
Top pattern	`top_fill_pattern`	`top_surface_pattern`
Infill angle	`fill_angle`	`infill_direction`
First layer height	`first_layer_height`	`initial_layer_print_height`
Line width	`extrusion_width`, `external_perimeter_extrusion_width`, …	`line_width`, `outer_wall_line_width`, `inner_wall_line_width`, …
Flow multiplier	`extrusion_multiplier`	`filament_flow_ratio`
Bridge flow	`bridge_flow_ratio`	`bridge_flow`
Retraction length	`retract_length`	`retraction_length`
Retract only above	`retract_before_travel`	`retraction_minimum_travel`
Lift	`retract_lift`	`z_hop` (+ `z_hop_types`)
Avoid crossing walls	`avoid_crossing_perimeters`	`reduce_crossing_wall`
Slow-down layer time	`slowdown_below_layer_time`	`slow_down_layer_time`
Fan layer time	`fan_below_layer_time`	`fan_cooling_layer_time`
Fan min / max	`min_fan_speed` / `max_fan_speed`	`fan_min_speed` / `fan_max_speed`
Support angle	`support_material_threshold`	`support_threshold_angle`
Purge volumes	`wiping_volumes_matrix`	`flush_volumes_matrix`

Concept	PrusaSlicer	Bambu Studio
Arc fitting	`arc_fitting`	`enable_arc_fitting`

19.2 Different defaults

The C++ defaults are fallbacks; the shipped profiles are what users get.

Setting	PrusaSlicer C++	PrusaSlicer "0.20mm SPEED @MK4S 0.4"	Bambu C++	Bambu base profile (`fdm_process_common.json`)
Layer height	0.3	0.2	0.2	0.2
Walls	3	2	2	2
Top layers	3	5	4	3 (5 in the 0.20 mm profiles)
Sparse infill	stars 20%	grid 15%	cubic 20%	grid 15%
Top pattern	monotonic	monotonic lines	rectilinear	monotonic line
Wall generator	Arachne	Arachne	Arachne	classic

Speeds in Bambu profiles are lists, e.g. `outer_wall_speed = ["200", "350"]` in "0.20mm Standard @BBL X1C": one value per extruder or nozzle variant.

19.3 Bambu printer features in the G-code

Most of these come from the printer profile templates in `resources/profiles/BBL/machine/`, not from C++. Some are written by `GCode.cpp` only when the printer is a Bambu printer (`is_BBL_Printer()`).

Command (Bambu firmware)	What Bambu Studio uses it for
`M620 S<n>A` … `M621 S<n>A`	Brackets an AMS filament change (in `change_filament_gcode`).
`M620.1`, `M620.11`	Flush feed rate/temperature and cutter retraction during a change.
`M1020 S<filament> H<nozzle>`	Tool change on Bambu multi-nozzle printers (`GCodeWriter::toolchange`) instead of `T<n>`.
`M73 L<layer>` and `M991 S0 P<layer>`	Layer progress and "layer changed" notification (`layer_change_gcode`).
`M971`, `M1004`, `M622`/`M623`	Timelapse photos (`time_lapse_gcode`), wrapped in `; SKIPPABLE_START/END`.
`M981 S1 P20000` / `M981 S0`	Spaghetti detector on / off.
`M1003 S1` / `M1003 S0`	Power-loss recovery on (from layer 2) / off.
`M976 S1 P1`	Scan the first layer with the lidar before layer 2 (`scan_first_layer`).
`M624` / `M625`	Marks the G-code of an object so the printer can skip it.

Command (Bambu firmware)	What Bambu Studio uses it for
`M1031 S1` / `M1031 S0`	Marks the start and end of ironing.

Bambu's change-filament template, for example, flushes with `F{flush_volumetric_speeds[...]/2.4053*60}`: mm³/s ÷ 2.4053 mm² = mm/s of filament, × 60 = mm/min.

19.4 The `.gcode.3mf` file

A Bambu printer is usually sent a `.gcode.3mf`, which is a ZIP archive. From `Format/bbs_3mf.cpp`, it holds, among others:

Path inside the archive	Contents
`Metadata/plate_1.gcode`	The G-code for plate 1 (`plate_N.gcode` for plate N).
`Metadata/plate_1.gcode.md5`	Checksum of that G-code.
`Metadata/plate_1.png`, `plate_no_light_1.png`	Thumbnails.
`Metadata/slice_info.config`	What was sliced: printer, filaments, time, weight.
`Metadata/model_settings.config`, `project_settings.config`	Object and project settings.
`3D/3dmodel.model`	The model itself (standard 3MF).

The G-code inside is split into commented blocks: `; HEADER_BLOCK_START … END` (time, layers, filament totals), `; CONFIG_BLOCK_START … END` (every setting) and `; EXECUTABLE_BLOCK_START … END` (the moves).

19.5 Other Bambu-only logic

- Shared-object slicing cache (section 6.3).
- `posDetectOverhangsForLift`: lifts the nozzle on travel over overhangs.
- `posSimplifyWall/Infill/SupportPath`: simplifies paths after generation, with extra simplify steps for supports.
- Bucketed overhang speeds and speed smoothing (16.5).
- Calibration G-code generators (`Calib.cpp`): pressure advance lines, flow tests.
- Multi-nozzle and filament grouping for H2-series printers (`FilamentGroup.cpp`, `MultiNozzleUtils.cpp`).

20. What PrusaSlicer does differently

- **Binary G-code** (`.bgcode`, `binary_gcode` option): a compact, checksummed binary format with blocks for metadata, thumbnails and G-code, used by newer Prusa printers.
- **Two time estimates** (normal and stealth `M73`).
- **Support spot detection** (`SupportSpotsGenerator.cpp`): warns the user when a part probably needs supports they did not enable.
- **Curled-extrusion estimation** (`posEstimateCurledExtrusions`): predicts where an overhang will curl up and slows down near it.
- **Dynamic, continuous overhang speeds and fan** (16.5).
- **Rewritten seam placer** with separate aligned, rear, random and scarf strategies.
- **MMU wipe tower** for Prusa's single-nozzle multi-material unit, with ramming and cooling moves.
- **Label objects** in Marlin/Prusa (`M486`) or Klipper (`EXCLUDE_OBJECT_*`) syntax (`GCode/LabelObjects.cpp`).

21. Side-by-side comparison

Area	PrusaSlicer 2.9.6	Bambu Studio 02.08.04.61
Origin	Slic3r → PrusaSlicer	Fork of PrusaSlicer
Licence	AGPL-3.0	AGPL-3.0
Integer scale	1 nm	10 nm
Layer heights	`generate_object_layers`	Same algorithm
Mesh slicing	`TriangleMeshSlicer`	Same, inherited
Walls	Classic or Arachne	Classic or Arachne
Surface classification	`detect_surfaces_type` etc.	Inherited, extended
Flow formula	Rounded rectangle	Identical
Overhang speed	Continuous interpolation by overhang distance	5 buckets by overhang fraction
Cooling	`CoolingBuffer`	Same formula, in `GCodeEditor`
Supports	Grid, snug, organic	Normal/tree × several styles, own tree support + organic
Output	`.gcode` or `.bgcode`	`.gcode` or `.gcode.3mf`
Printer-specific G-code	Profiles (`PrusaResearch.ini`)	Profiles (`resources/profiles/BBL/`) + `is_BBL_Printer()` code
Identical objects	Sliced per object	Sliced once, copied

22. What this means for Caliper

22.1 Scope

`docs/design.md` currently lists "slicing and G-code" as a **non-goal**. Adding a slicer changes that, so the first implementation PR should come with an ADR in `docs/adr/` that records the decision, the alternatives and the licence choice below.

22.2 The licence question: read this before writing code

Both slicers are **AGPL-3.0**. Caliper is **MIT**. In plain terms (this is not legal advice; check with someone qualified before shipping):

- **Ideas, algorithms and formulas are not protected by copyright; code is.** Implementing "cut each triangle at mid-layer height and chain the segments by shared edges" or "area = h × (w − h × (1 − π/4))" in our own TypeScript, in our own structure, is reimplementation.
- **Translating their C++ into TypeScript line by line, or copying it, makes Caliper a derivative work** that must then be released under the AGPL, and the AGPL's network clause applies to a web app.
- **Compiling their code to WebAssembly and shipping it with Caliper** also puts that part under the AGPL, with the same consequence for the combined app.

So there are three paths:

Option	What it means	Trade-off
A. Reimplement (recommended)	Write our own slicer from this report and the general literature, matching the *behaviour* and maths, not the code. Keep MIT.	Most work; results will differ in detail from the real slicers.
B. Relicense Caliper to AGPL-3.0	Then porting or compiling their code is allowed.	Changes the project's licence for everyone; all contributors must agree.
C. Don't slice in Caliper	Export a ready-to-slice 3MF/STL and open it in the user's slicer.	No G-code in Caliper.

"Adding the exact logic", under option A, should mean **the same pipeline, rules and formulas**, verified by comparing outputs, not the same source code.

The libraries are a separate question. Both slicers use **Clipper** for polygon operations, and Clipper/Clipper2 are under the **Boost Software License**, which is compatible with MIT, so a JavaScript or WebAssembly build of Clipper2 can be used.

22.3 How it fits Caliper's architecture

- **Keep `viewer/` and analysis code free of React**, as today. A slicer is pure computation: it belongs next to `printability.ts`, written against `TriangleSource` (`src/viewer/triangles.ts`), so it runs in Node's test runner too.
- **Run it in a Web Worker.** Slicing a real model takes seconds; on the main thread it would freeze the UI.
- **Axis conversion.** Caliper uses three.js, which is **Y-up** (X right, Y up, +Z towards the viewer). Slicers and printers are **Z-up** (X right, Y away from the viewer, Z up). Converting a Caliper point to printer space: `printer = (x, −z, y)`, then shift so the bed's front-left corner is (0, 0).
- **Units.** Caliper already converts file units to millimetres (`MM_PER` in `src/lib/printers.ts`). Slice in millimetres, scaled to integers for Clipper.
- **Reuse what exists.** The Preview tab's layer slider (`layerCount`, `layerAtFraction` in `printers.ts`) and the overhang check already follow the same ideas as steps 1 and 7.

22.4 A phased plan

Each phase ends with something visible and testable.

Phase	Build	Prove it with
1. Layers	`generate_object_layers` equivalent: fixed then variable heights.	Unit tests with the worked example in 7.4.
2. Outlines	Cut triangles at `slice_z`, chain by shared edges, close gaps, build ExPolygons with the non-zero rule. Draw them in the Preview tab.	A cube, a cylinder and a hollow box give the expected areas per layer; a mesh with a hole still closes.
3. Walls	Classic offset walls only.	Wall count and spacing match the formulas; draw walls in Preview.
4. Surfaces	Top/bottom detection, solid shells, sparse area.	A box with 3 top layers has exactly 3 solid layers under its top.
5. Infill	Rectilinear and monotonic solid, grid sparse, density → spacing.	Line spacing equals `spacing / density`.
6. Flow and G-code	Flow maths, a G-code writer for one printer (Bambu P1S or A1), start/end templates, travel with retraction.	Total extruded volume ≈ part volume × fill (Caliper already computes part volume); the file opens in the real slicer's G-code viewer.
7. Cooling and speeds	Per-role speeds, volumetric cap, cooling slow-down and fan.	The worked examples in 15.5 and 17.4.
8. Supports, brim, skirt	Grid supports first; trees much later.	Visual checks; prints.

Arachne, tree supports, seam optimisation and multi-material are advanced work. They should come only after the simple version prints correctly.

22.5 How to test against the real slicers

- **Golden comparisons.** Slice the same small models (a 20 mm cube, a cylinder, a bridge test) in PrusaSlicer (it has a command-line mode, e.g. `prusa-slicer --export-gcode model.stl`) and Bambu Studio, with the same settings, and compare per layer: number of layers, total extrusion length per role, number of retractions. Exact line-by-line equality is not the goal; matching totals within a tolerance is.
- **Invariants.** Extruded volume ≈ printed volume; no move outside the bed; Z never goes down within one object; every `G1` with `E` > 0 is inside the layer's outline grown by half a line width.
- **Print it.** The final check is a real print of a calibration cube.

23. Glossary

Term	Meaning
AMS	Bambu's Automatic Material System, which switches between up to four filaments.
Arachne	Variable-width wall generator from CuraEngine.
Bridge	Extrusion across empty space, held only at its ends.
Clipper	The polygon boolean/offset library both slicers use.
Contact distance	Gap between support and part.
Elephant foot	The first layer spreading wider than designed.
E	Extruder position or distance in G-code (mm of filament).
ExPolygon	A contour polygon with hole polygons.
Flow ratio / extrusion multiplier	Correction factor on extruded amount.
Gap fill	Thin lines filling slivers between walls.
Ironing	Re-passing a top surface with tiny flow to smooth it.
Layer height profile	List of (z, layer height) points defining variable layer heights.
Manifold	A closed mesh where every edge belongs to exactly two triangles.
Medial axis	The set of points equally far from two or more sides of a shape; Arachne builds walls along it.
Monotonic	Infill printed so every line goes in the same direction relative to its neighbour.
Morphological opening / closing	Shrink-then-grow (removes slivers) / grow-then-shrink (closes cracks).
Non-zero rule	Fill rule: a point is inside if the outlines wind around it a non-zero number of times.
Offset	Growing or shrinking a polygon by a distance.
Ooze	Molten plastic leaking from the nozzle during travel.
Placeholder	`[name]` or `{expression}` in custom G-code, filled in by the slicer.
Pressure advance	Firmware feature that pushes extra filament at the start of a line and less at the end.
Region	Part of a layer sharing one set of settings.
Retraction	Pulling filament back before travel.
Scarf seam	A seam where the loop's start and end overlap on a slope.
Seam	Where a wall loop starts and ends.

Term	Meaning
Shell	Walls plus solid top/bottom layers.
Skirt / brim	Loops around (skirt) or attached to (brim) the first layer.
slice_z / print_z	The cutting height (middle of the layer) / the top of the layer.
TBB	Intel Threading Building Blocks, the parallel-computing library both use.
Volumetric speed	mm^3 of plastic per second; the hot end has a maximum.
Wipe tower / prime tower	A tower used to purge filament after a change.
Z-hop	Lifting the nozzle during travel.

24. Sources and how to check them yourself

Every statement above was read from the source releases named in section 1. To look for yourself without downloading the whole repositories:

```
# PrusaSlicer 2.9.6, only the geometry library
git clone --depth 1 --filter=blob:none --sparse --branch version_2.9.6 \
  https://github.com/prusa3d/PrusaSlicer.git
cd PrusaSlicer && git sparse-checkout set src/libslic3r resources/profiles

# Bambu Studio 02.08.04.61, the geometry library and Bambu printer profiles
git clone --depth 1 --filter=blob:none --sparse --branch v02.08.04.61 \
  https://github.com/bambulab/BambuStudio.git
cd BambuStudio && git sparse-checkout set src/libslic3r \
  resources/profiles/BBL/machine resources/profiles/BBL/process
```

The files worth reading, in the order of this report:

Topic	**File(s) in** `src/libslic3r/`
Step list and `Print::process`	`Print.hpp`, `Print.cpp`
Layer heights	`Slicing.cpp` (`generate_object_layers`), `SlicingAdaptive.cpp`
Mesh slicing	`TriangleMeshSlicer.hpp/.cpp`, `PrintObjectSlice.cpp`
Elephant foot	`ElephantFootCompensation.cpp`
Walls	`PerimeterGenerator.cpp`, `Arachne/`
Surface types	`Surface.hpp`, `PrintObject.cpp` (`prepare_infill` and its sub-steps)
Bridges	`BridgeDetector.cpp`
Infill	`Fill/Fill.cpp`, `Fill/FillBase.hpp`, `Fill/FillRectilinear.cpp`, `Fill/FillGyroid.cpp`
Supports	`Support/SupportMaterial.cpp`, `Support/TreeSupport*.cpp`
Flow	`Flow.hpp/.cpp`, `Extruder.hpp/.cpp`
G-code export	`GCode.cpp` (`_do_export`, `process_layers`, `process_layer`, `_extrude`, `needs_retraction`)
G-code text	`GCodeWriter.cpp` (PrusaSlicer: `GCode/GCodeWriter.cpp`)
Overhang speeds	PrusaSlicer `GCode/ExtrusionProcessor.cpp`; Bambu `GCode.cpp` (`get_overhang_degree_corr_speed`)
Cooling	`GCode/CoolingBuffer.cpp`; Bambu also `GCode/GCodeEditor.cpp`
Time estimate	`GCode/GCodeProcessor.cpp`

Topic	File(s) in `src/libslic3r/`
Settings and defaults	`PrintConfig.hpp/.cpp`
Bambu 3MF output	`Format/bbs_3mf.cpp`
Bambu printer templates	`resources/profiles/BBL/machine/*template*.json`

A suggested way to learn from it. Pick one small function, for example `Flow::mm3_per_mm` or `generate_object_layers`. Read it, write the same thing in TypeScript from memory without looking, write a test with numbers you worked out by hand, then compare with the original. That builds real understanding, and it keeps our code our own (section 22.2).

25. Caliper's slicer: what was built

Section 22 planned the work; this section records what exists, so the report and the code can be read side by side. The licence path is option A (22.2): the pipeline below is Caliper's own TypeScript, written from this report, using Clipper 6 (`clipper-lib`, Boost licence) for polygon operations. Nothing is copied or translated from either slicer.

25.1 Where each step lives

Step in this report	Caliper file	What it does	Differences from the slicers
Data model, integer coordinates (5.5)	`src/slicer/clipper.ts`	ExPolygons, offsets, booleans, line clipping; 1 unit = 1 µm	µm instead of nm (PrusaSlicer) or 10 nm (Bambu)
Layer heights (7)	`src/slicer/layers.ts`	`generateLayers`: fixed first layer, then layers cut at their middle	No variable or adaptive layer height yet
Mesh slicing (8)	`src/slicer/slice.ts`	Weld vertices, cut triangles, chain by shared edges, close gaps ≤ 1 mm, non-zero union, 0.05 mm closing	Same idea; simpler gap closing
Walls (9.3)	`src/slicer/walls.ts`	Classic walls: outer at w/2, then (spacing + spacing)/2, then spacing; infill 25 % into the last wall	No Arachne, no gap fill, no thin walls (reported instead)
Surfaces (10)	`src/slicer/surfaces.ts`	`sparse = infill ∩ T layers above ∩ B layers below`; the rest is solid, split into top, bridge and internal	One rule covers horizontal and vertical shells
Infill (11)	`src/slicer/infill.ts`	Lines on a bed-fixed grid, alternating direction; grid and lines patterns; bridge angle = shortest span	No gyroid, cubic, lightning
Supports (13)	`src/slicer/surfaces.ts → overhangs`	**Not generated.** Overhangs past the limit and islands starting in mid-air are reported	Mid-air islands block export
Skirt, brim (14)	`src/slicer/toolpaths.ts`	Skirt around everything, brim loops from the outside in	No draft shield, no brim ears
Toolpath order	`src/slicer/toolpaths.ts`	Islands nearest-first; inner walls before outer; seam at the rear or aligned; dense fills joined in a zig-zag; fill pieces nearest-first	—
Flow maths (15)	`src/slicer/flow.ts`	The rounded-rectangle formulas, bridge circle, E per mm³, volumetric cap	Identical formulas
G-code writing (16)	`src/slicer/gcode.ts`	Relative E, per-role speeds, retraction only when a travel crosses a wall, Z-lift, purge line, templates with a fixed placeholder list	No arc fitting, no pressure advance

Step in this report	Caliper file	What it does	Differences from the slicers
Cooling (17.4)	`src/slicer/gcode.ts → fanFor, slowdownFactor`	PrusaSlicer's fan formula and minimum layer time	Applied per layer before writing, not as a filter
Time estimate (17.5)	`src/slicer/motion.ts`	Look-ahead trapezoid planner with Marlin's junction deviation	—
Orchestration	`src/slicer/index.ts`	`slicePlate`: refuses bad input, runs the steps, verifies, merges findings	—
Worker	`src/slicer/worker.ts`	Runs slicing and checking off the main thread	—
Printer, filament, process data	`src/slicer/profiles.ts`	Limits, temperatures, speeds, start and end sequences	Start sequences are Caliper's own

25.2 Printers

Printer	Start sequence	Notes
Creality Ender-3	Built in: G28 then heat, hand-levelled bed	Nozzle capped at 245 °C for the PTFE-lined hot end; Z at 5 mm/s
Prusa MK4	Built in: model and nozzle checks (M862), probe at 170 °C, G28, G29 mesh	Experimental
Bambu Lab X1/P1/A1, A1 mini	**You supply it**	Bambu's own sequence wipes the nozzle, levels, and drives the AMS; Caliper does not imitate it

26. Verifying G-code before it reaches a printer

A wrong G-code file can damage a printer: a nozzle driven into the bed, a hot end heated past its rating, filament ground against a cold nozzle, a blob the nozzle crashes into. Caliper never assumes its own output is right. Every file is **read back as text**, the way the printer will read it, and checked against the chosen printer's limits (`src/slicer/verify.ts`). The Preview draws the moves that check collected, so what you look at is the file itself.

26.1 What is checked

Errors block export. Warnings must be acknowledged. For files from other slicers, a few errors become warnings, because vendor start sequences do things Caliper cannot judge (marked "softer for other slicers").

Code	Check	Why it matters	Severity
E001	Every number is a finite number	A garbled value can be read as a huge move	Error
E002	No unfilled `{placeholder}` or `[variable]`	Usually a start sequence pasted from another slicer	Error
W003	Every line is a G, M or T command	Macros cannot be checked	Error (warning for other slicers)
E010	No axis moves before G28 homes it, or after motors are released	The printer does not know where the nozzle is	Error
E011	Every move inside the machine's travel, arcs included	Grinding against end stops or the frame	Error
E012	Plastic only on the bed	Extruding into the air or the frame	Error (warning for other slicers' purge lines)
E020, E021	Nozzle never below the bed; never extruding below 0.05 mm	Scraping or gouging the bed	Error
E022	Nothing printed higher than the printer can	Collision with the gantry	Error
E023, E024	Never printing or travelling below a finished layer	The nozzle ploughs through the part	Error
E030, E031	Nozzle and bed temperatures within the printer's rating	PTFE fumes, damaged hot end or bed	Error
E034, E035	No extrusion while the nozzle is under 170 °C or before an M109 wait	Ground filament, jammed extruder	Error
W036	The bed is waited for before printing	Poor adhesion	Warning

Code	Check	Why it matters	Severity
W032	Printing temperature within the filament's range	Poor layers, clogging	Warning
E040	Every move has a non-zero speed	Stalled moves	Error
W041, W042, W044	Speeds and accelerations within the firmware's limits	The firmware caps them; times are off	Warning
W050	Flow within what the hot end melts	Under-extrusion	Warning
E051	No line wider than 2.5 × the nozzle	A ridge the nozzle collides with	Error
W052, E053	Retractions within the extruder's safe length	Molten plastic pulled into the cold zone jams	Warning, error past 3 ×
E054	No large blob pushed out in one place	The nozzle hits the blob	Error
E060	No tool change on a single-nozzle printer	Undefined behaviour	Error
E061	Fan values 0–255	Invalid command	Error
W070, W071	Heaters off at the end; nozzle lifted off the print	Fire risk, melted part	Warning
E080	The file was made for the printer chosen	Its limits would not apply	Error
E081, E082, E083	Caliper's own files: the declared extrusion total matches, the start markers and end-of-file marker are present	The file was edited or cut short after slicing	Error
E100	Something is printed at all	An empty file	Error
S001–S004	The slicer's own findings: open mesh, features thinner than a line, overhangs past the limit, parts starting in mid-air	A failed print, or loose plastic wrapping the nozzle	Warning; mid-air is an error

Every check has a test that feeds the checker a file with exactly that fault (`test/verify.test.ts`), and a test that a correct file passes. The slicer's own maths is tested against the worked examples in this report (`test/slicer.test.ts`, `test/motion.test.ts`).

26.2 Before the first print: a procedure

The checker can only see the file. These steps cover the printer.

1. **Slice in Caliper and read the report.** Fix every error. Read every warning; the usual fix is to reorient the part or change a setting.
2. **Step through the Preview.** Drag the layer slider to layer 1 and look at the first layer: it should sit inside the bed, with the skirt around it. Turn on travel moves for a layer or two.

3. **Cross-check in a second viewer.** Open the exported file in a G-code viewer you already trust, for example PrusaSlicer's G-code viewer (File → G-code preview) or your printer's own preview. The layers, the time and the filament length should agree roughly with Caliper's.
4. **Check a known-good file the same way.** Use "Check a G-code file" on a file from your usual slicer for the same printer. Comparing the two reports shows whether Caliper's limits match your printer.
5. **Start small.** Print a 20 mm cube first, with the bed clean, the right filament loaded and, on printers without automatic levelling, the bed levelled.
6. **Watch the first layer with a hand near the stop control.** Stop the print if the nozzle digs in, the plastic does not stick, or anything moves where it should not. On most printers this is a "Stop" or "Cancel" item on the screen; cutting the power is the last resort.
7. **Only then print larger parts,** and keep watching the first layer of each new kind of part.